# Experimental evidence on the benefits of the geometric offload under growing workload

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## 1 Motivation and objectives

The thesis showed that the **offload** redistributes the computational load without appreciably altering the final reconstruction. It remains to be established whether the capacity so recovered at the compression unit yields measurable benefits as the requested rate grows. To that end, the two placements are compared both at the **nominal cadence** and beyond the sustainable capacity of the chain. The repetitions make it possible to separate systematic differences from the variability between executions and to examine the response of the **temporal regulator**.

## 2 Configuration of the runs and criteria of analysis

The experiment comprises **20 runs**, divided equally between the **B0** and **O-OFF** placements and the requested rates of 30 **and** 40 **frame / s**, with five repetitions for each combination. Every execution processes the 300 frames of the **Loot** sequence, on the same host and with an unchanged compilation of the program. The interactive mode described in the thesis is retained, with the fidelity capture disabled. The analysis follows the methodological criteria already adopted: the stage measurements are confined to the **steady window** $10 \leq \texttt{frame_id} \leq 285$, whereas the terminal indicators admit the **complete** frames alone. The percentiles are obtained by linear interpolation and the effective cadence derives from the temporal span of the admitted frames. For each combination the median of every repetition is determined, and the central value of the five estimates is then taken. Differences between the groups are assessed with the **two-sided Mann-Whitney test**, whereas binary outcomes are examined through **Fisher's exact test**. At this sample size, both return $p \simeq 0.0079$ under **complete separation**.

## 3 Consistency with the original document

The results confirm what the thesis observed for every directly comparable measure. The **times of the geometric stages** that the **O-OFF** configuration returns to the **Encoder** amount to 4.800 and 2.348 ms, against the intervals 4.761-5.109 and 2.391-2.411 ms recorded earlier. The **population-dependent work** of the compression unit, defined further on, measures 8.823 ms in the nominal placement, where the dissertation reports 9.466 ms, and 17.292 ms in the ablation, compared with 17.140-18.104 ms. Taking every complete frame rather than the steady window alone, the **source-terminal delay** amounts to 212.012 ms, against the 216.104 ms obtained previously. Measured on the reference pipeline, which records 337.599 ms, the nominal placement attains a **reduction of** 37.20%, where the thesis reports 35.99%. A difference emerges instead in the **transmission time of the source**, comprised between 7.085 and 7.261 ms in four of the five nominal executions, as opposed to the 11.965 ms measured earlier. The state of the coder follows that variation rather than the placement of the geometric stages. Section 8 examines this **nuisance factor**, to be treated separately rather than aggregated through a mean.

## 4 The cost the offload removes

Within the Encoder, four stages contribute to the cost that grows with the number of admitted points: the conversion of the incoming coordinates, the aggregation of the geometric descriptor, the computation of the associated radius and the projection onto the atlas. Their sum defines the **population-dependent work**, which the placement of the geometric stages governs directly. The remaining component of the host interval is the submission of the picture to the coder, performed on an atlas of fixed size and therefore not proportional to the number of points.

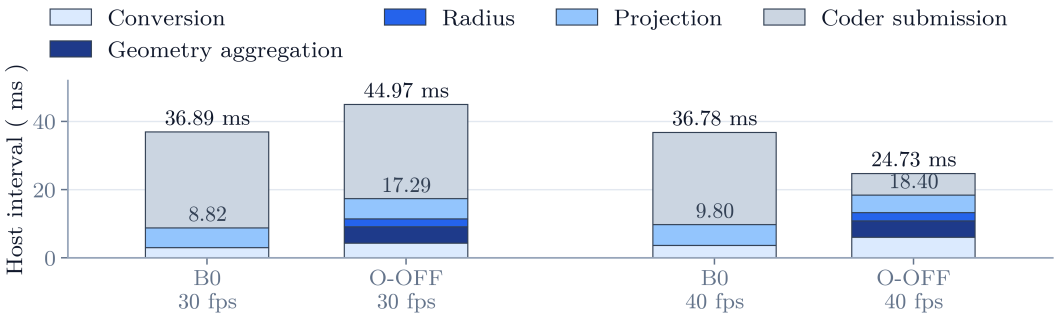
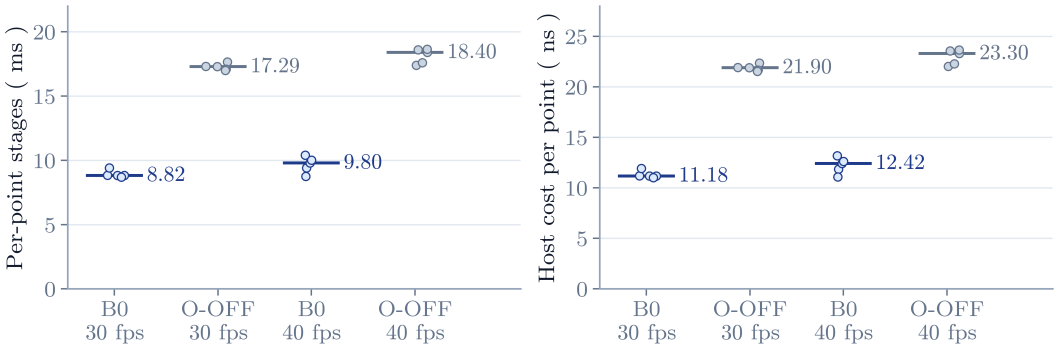



**Figure 1:** Decomposition of the host interval of the compression unit. The values inside the columns give the population-dependent work

In the **O-OFF** configuration, the two geometric stages require 7.148 ms in all at 30 frame / s and 7.242 ms at 40 frame / s, with a variation held within 0.3 ms across the ten executions. The conversion adds a further 1.275 and 2.369 ms over B0, since the unit has to materialise the whole population before traversing it. Taken as a whole, the population-dependent work turns out **1.96 times larger** at the nominal cadence and **1.88 times** at the elevated one, without overlap between configurations.

| **Quantity** | **B0** 30 | **O-OFF** 30 | $p$ | **B0** 40 | **O-OFF** 40 | $p$ |
|---|---|---|---|---|---|---|
| Conversion | 3.006 | 4.282 | 0.0159 | 3.654 | 6.023 | 0.0079 |
| Geometry aggregation | 0.000 | 4.800 | 0.0079 | 0.000 | 4.795 | 0.0079 |
| Radius | 0.000 | 2.348 | 0.0079 | 0.000 | 2.447 | 0.0079 |
| Projection | 5.739 | 5.926 | 0.4206 | 6.071 | 5.109 | 0.0159 |
| Population-dependent work | 8.823 | 17.292 | 0.0079 | 9.803 | 18.395 | 0.0079 |
| Host cost per point | 11.175 | 21.901 | 0.0079 | 12.420 | 23.302 | 0.0079 |
| Coder submission | 28.143 | 27.611 | 0.7937 | 27.052 | 6.353 | 0.0079 |
| Active interval | 37.549 | 47.185 | 0.4206 | 38.768 | 25.319 | 0.0317 |

**Table 1:** Stage times of the compression unit and statistical significance. Values in milliseconds, except the cost per point expressed in nanoseconds




**(a)** Population-dependent work **(b)** Cost per admitted point

**Figure 2:** Values of the single executions, with the median of the five repetitions marked by a bar. The two configurations occupy distinct intervals at both cadences

Referred to the roughly 789 thousand admitted points, the **host cost per point** amounts to 11.175 ns in B0 and 21.901 ns in O-OFF at the nominal cadence. Under the conditions observed, the placement without the offload demands work per point **almost twice as large** as the configuration that transfers the geometric stages.

## 5 From the recovered time to the reserve of resolution

The **geometric offload** does not reduce the cost of the geometric stages, but changes where in time they fall. The ingress node requires 10.637 ms when it computes

the descriptor and 2.040 ms in its absence, an increment of 8.597 ms, practically equivalent to the 8.469 ms the compression unit saves. The difference lies in the possibility of advancing them progressively, as the packets arrive, completing them when the last sample lands. Without the offload, the computation begins on a whole frame and adds itself in series to the following stages. The sum of the active intervals over the nine nodes amounts to 107.591 ms in **B0** and 114.154 ms in **O-OFF**.

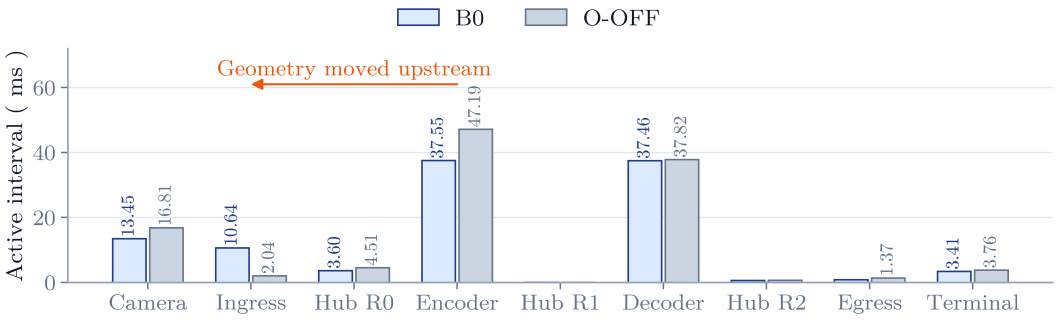
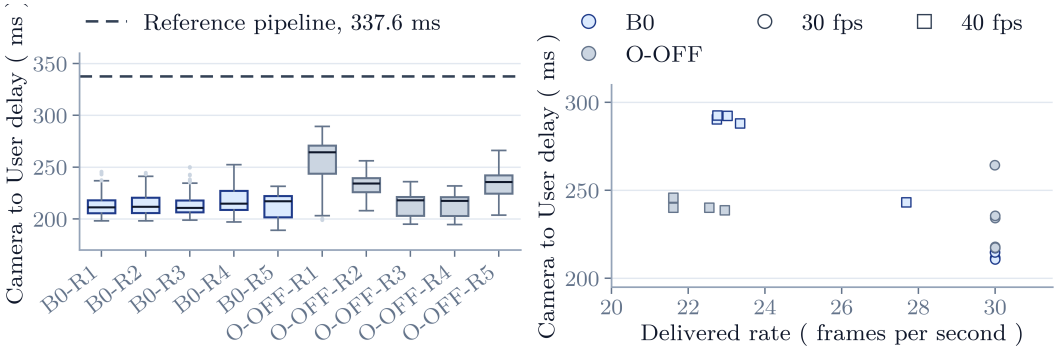



**Figure 3:** Distribution of the active intervals over the nine nodes at 30 frame / s

The time recovered at the compression unit constitutes a **reserve of resolution**, which the host cost per point makes it possible to quantify. Assuming a linear growth of the four population-dependent stages, all else being equal, the largest number of points that can be traversed within the period of 33.333 ms follows from dividing that interval by the unit cost. The theoretical capacity thus reaches about 2.98 **million points per frame** in B0, against 1.52 million in O-OFF. Setting aside for the submission to the coder the 8-9 ms observed in its relaxed state, the respective limits fall to about 2.27 and 1.11 million. Related to the 0.790 million points of the sequence employed, this corresponds to a capacity of nearly three times the present density with the offload, against roughly 1.4 times without it. The same relation permits an estimate of the effect of a **doubling of the population**: the work of the four stages would pass from 8.823 to 17.65 ms in B0 and from 17.292 to 34.58 ms in O-OFF, exceeding the available period in the latter case. The benefit therefore concerns not only the time of the computation, but also the **capacity to sustain larger resolutions**. These values nevertheless descend from a linear extrapolation of the measured costs and not from runs conducted at different densities.

## 6 Delay and stability at the nominal cadence

At 30 frame / s, both configurations admit and deliver all 300 frames, which permits a comparison at equal load and requested cadence. The **source-terminal delay** amounts to 211.666 ms in B0 and 234.248 ms in O-OFF, with a **reduction of** 22.582 **ms** in favour of the offload. The medians of the five repetitions occupy the intervals 210.743-217.168 ms and 217.442-264.352 ms respectively.




**(a)** Delay per frame at 30 frame / s **(b)** Delay and effective cadence

**Figure 4:** Source-terminal delay in the B0 and O-OFF configurations

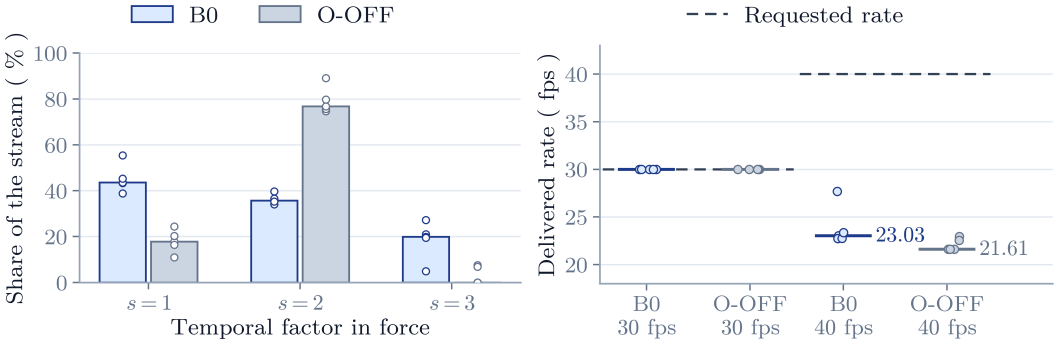
This result clarifies an uncertainty that arose in the thesis, where the direction of the benefit was reproducible but its extent was not: two repetitions without the offload showed differences from the nominal configuration of 11.81 and 0.15 ms. With five executions per group, the advantage is confirmed and attains **statistical significance** at $p \simeq 0.0079$, the smallest value the adopted test can return. The **temporal stability** demands instead that two levels of variability be distinguished. Between executions, B0 presents a dispersion of the five medians of 6.43 ms, against 46.91 ms in O-OFF. The gap persists even when only the runs with a congested coder are considered, with intervals of 3.98 and 30.10 ms respectively. Within the single repetitions, by contrast, the differences are less pronounced: the interquartile range amounts to 14.77 against 18.30 ms ( $p = 0.3095$ ), whereas the standard deviation equals 11.25 against 10.32 ms. The distance between the median and the ninety-fifth percentile turns out even smaller in O-OFF, at 14.42 against 23.27 ms. The benefit therefore bears on the **reproducibility between executions**, not on a greater regularity inside the individual repetitions.

## 7 Response of the regulator to an unsustainable cadence

At 40 frame / s, neither configuration reaches the requested cadence. The **temporal regulator** reduces the admitted frames in all ten runs, bringing B0 and O-OFF to distinct operating points without interrupting the processing.

| Quantity at the elevated request | B0 | O-OFF | $p$ |
|---|---|---|---|
| Frames admitted of three hundred | 172 | 162 | 0.0397 |
| Cadence delivered in frames per second | 23.026 | 21.611 | 0.0317 |
| Share of the stream at the full factor | 43.6 % | 17.8 % | 0.0079 |
| Share of the stream at the halved factor | 35.7 % | 76.7 % | 0.0079 |
| Smoothed interval of the accelerator | 9.874 | 12.752 | 0.0079 |
| Frames below the release threshold | 57.7 % | 35.6 % | 0.0079 |

**Table 2:** ndicators of the regulator at 40 frame / s, with the thinning active, reported as the median of the five repetitions for each configuration




**(a)** Partition of the stream by factor **(b)** Requested and effective cadence

**Figure 5:** Behaviour of the two configurations at the nominal and elevated cadences

With the offload, the **full factor** is held for 43.6 % of the stream, against 17.8 % in O-OFF, with no overlap between the five repetitions of the two groups. The cadence delivered amounts to 23.026 and 21.611 frame / s respectively. The difference does not depend on an earlier onset of the thinning. The first rise of the factor occurs at frame 15 in B0 and at 17 in O-OFF ( $p = 0.7619$ ), while both configurations leave the warm-up phase of the regulator at the same moment. The first recovery likewise requires 56 frames from the initial rise in either case. What changes is the **duration of the sojourns**: above the full factor, the mean one lasts 17.55 frames with the offload and 46.20 without, whereas the longest reaches 31 and 77 frames respectively. At the full factor the chain remains on average for 13.60 frames in B0, against 6.50 in O-OFF. The first configuration therefore records 15 transitions against the 9 of the second, alternating the levels of thinning more often, yet holding them for shorter stretches throughout the observed stream.

| Dynamics of the temporal factor | B0 | O-OFF | $p$ |
|---|---|---|---|
| Frame of the first rise | 15 | 17 | 0.7619 |
| Frames to the first return to the full factor | 56 | 56 | 0.8571 |
| Mean sojourn above the full factor | 17.55 | 46.20 | 0.0079 |
| Longest sojourn above the full factor | 31 | 77 | 0.0079 |
| Mean sojourn at the full factor | 13.60 | 6.50 | 0.0079 |
| Transitions along the stream | 15 | 9 | 0.0079 |

**Table 3:** Sojourns and transitions of the regulator at 40 frame / s, expressed in frames

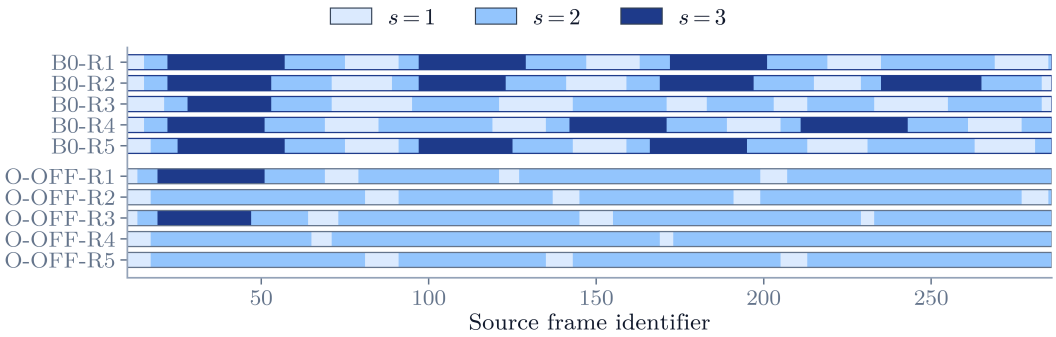
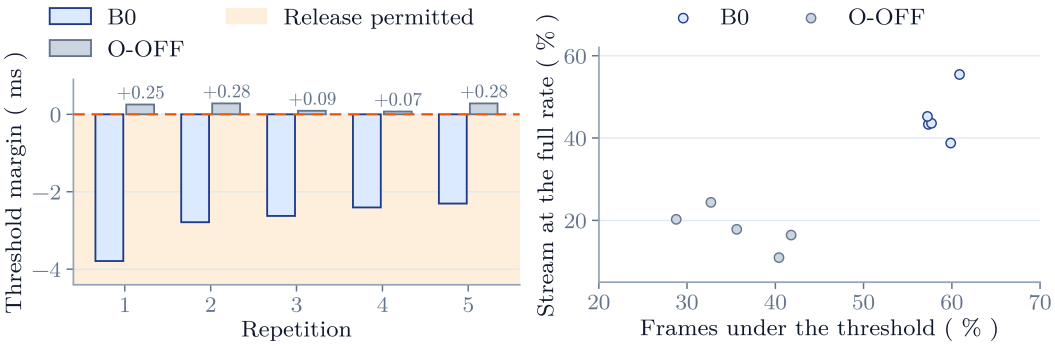



**Figure 6:** Evolution of the temporal factor over the ten executions at the elevated request

This behaviour depends on the **release threshold** of the regulator. The loop applies an exponentially weighted moving average to the interval of the accelerator stage, reconstructible from the times of projection, geometric aggregation, radius computation and readiness wait, with a largest deviation of 0.022 ms from the recorded values. That correspondence identifies the components the control observes, and excludes the coder. A reduction of the factor requires the ratio projected at the level immediately below to remain under 0.5, a condition which, at a base period of 25 ms, amounts to a smoothed interval below 12.5 **ms**. B0 presents values comprised between 8.709 and 10.196 ms, satisfying the criterion in all five repetitions, whereas O-OFF records 12.572-12.783 ms, exceeding the limit every time.




**(a)** Smoothed interval and release threshold **(b)** Release condition and share of the stream

**Figure 7:** Release conditions and response of the regulator at 40 frame / s

The roughly 7 ms that the offload transfers therefore let B0 settle below a **structural threshold**, while O-OFF remains beyond the limit. The result does not concern a difference between means alone: the regulator compares a quantity directly governed by the geometric placement with a value fixed by the control algorithm. Two distinctions nevertheless remain important. The rise of the factor is not determined by the projected ratio, which at the full level reaches 0.35-0.41 in B0 and 0.50-0.51 in O-OFF, staying under the threshold of 0.75. The increase depends instead on the depth of the **coder** queue and on the attendant waits: it is this stage, not the geometry, that prevents 40 frame / s from being reached. Moreover, since the configurations attain different effective cadences, their respective delays are not directly comparable. B0 records 290.386 ms at about 23.0 frame / s, while O-OFF measures 240.172 ms at 21.6 frame / s. These values describe different points of the **trade between cadence and delay**, not an isolated worsening of one configuration. Figure 4b represents the two requests through four distinct groups of observed operating points for the two configurations.

## 8 The state of the coder

The submission of the picture to the **coder** presents two distinct regimes: 8.020-9.018 ms in three of the ten nominal runs and 27.611-33.885 ms in the remaining seven. The slower regime appears in four of the five repetitions of B0 and in three of the five of O-OFF, with no statistically detectable association with the placement ( Fisher's exact test, $p = 1.0$ ). What emerges instead is a **negative correlation** with the duration of the transmission of a frame by the source, equal to $r = -0.842$ over the ten executions. This variability carries two consequences. First of all, the overall active interval of the compression unit does not isolate the geometric contribution, since the effect of the coder prevails over the difference between the placements ( $p = 0.4206$ at the nominal cadence ). The **population-dependent work** of Table 1 preserves instead a complete separation. Furthermore, the state of the coder governs the extent of the terminal benefit: in the congested regime the delay measures 211.428 ms in B0 and 235.553 ms in O-OFF, a gap of about 24.1 ms, whereas in the relaxed one the values are 217.168 and 217.672 ms, barely 0.5 ms apart. The direction of the advantage stays unaltered, but its amplitude depends on the interaction between the additional work and the congestion of the coder. The data collected do not include the clock and power state of the graphics device, the effective placement of the child process of the coder, nor an absolute temporal reference from which to reconstruct the order in which the runs were started. The **bimodality** can therefore be observed and held apart in the analysis presented here, without attributing a specific underlying cause to it.

## 9 Conditions of validity of the offload

The **geometric offload** transfers host work from the compression unit to the ingress node, without reducing its overall cost. The benefit thus depends on two conditions: that the compression be a limiting stage and that the ingress retain residual capacity. Both can be weighed through the measurements collected. With the 10.637 ms per frame reported in Section 5, the ingress occupies 31.9 % of the period at 30 frame / s and would occupy 42.5 % of it at a sustained cadence of 40 frame / s. A single stream therefore leaves a substantial margin, which renders the configuration examined a favourable one. Were several streams to share a single ingress core, and assuming a linear growth of the work, the **theoretical limit** would stand at about three concurrent streams at the nominal cadence and two at the elevated one, against about sixteen and twelve when the descriptor is computed at the compression unit. This extrapolation, unverified experimentally, identifies the conditions under which the ingress could become the new bottleneck. The second constraint already surfaces in the runs at 40 frame / s, where the **coder** prevents either configuration from sustaining the request. Since the submission operates on an atlas of fixed size, transferring the geometric stages does not reduce its cost directly. The offload can improve the regulator's response, yet not remove the limit this stage imposes.

## 10 Results and limits of the evidence

The twenty runs support **four principal conclusions**. In the first place, O-OFF presents a **population-dependent work almost twice as large** as that of B0, with ratios of 1.96 at the nominal cadence and 1.88 at the elevated one. The separation between the five repetitions of the two groups is complete in either case ( $p \simeq 0.0079$ ). In the second place, the cost of the geometric stages proves comparable at the two nodes: 8.597 ms added at the ingress against 8.469 ms withdrawn from the compression. The offload therefore brings the computation forward into the reception of the packets, rather than making it less onerous, and establishes a **theoretical capacity in points about twice** that of O-OFF, under the linear projection adopted. At the **nominal cadence**, B0 lowers the source-terminal delay by 22.582 ms with respect to O-OFF, with no overlap between the medians of the repetitions. The width of the interval those medians occupy is moreover about seven times smaller, which indicates a greater reproducibility between executions, not a reduced variability among the frames of one. Finally, at the request of 40 frame / s, the offload keeps the smoothed interval of the accelerator below the **release threshold** in five repetitions out of five, against none in O-OFF. B0 thus preserves the full factor for 43.6 % of the stream, as opposed to 17.8 %, with mean sojourns in the thinning shorter than half. These results do not, however, warrant four

further conclusions. First, O-OFF does not interrupt the processing at the elevated request, since the coder bounds both configurations. The delays observed in that condition do not permit a direct comparison either, being associated with different effective cadences. The **fidelity of the reconstruction** has not been reassessed in the present runs: its substantial neutrality remains a result of the thesis, with no new evidence of improvement. Lastly, the overall active interval of the compression does not furnish a dependable indicator of the benefit: B0-R1 records 37.549 ms against the 26.237 ms of O-OFF-R3, an inversion ascribable to the state of the coder described in Section 8. In the population-dependent work, by contrast, none of the twenty executions presents an inversion of that kind.